Contents

OpenPLC **1**
- Purpose . . . 1
- Role in the Architecture . . . 1
- Responsibilities . . . 1
- PLC Logic . . . 1
- Testing . . . 1
- Security-Lab Role . . . 2
- Related Components . . . 2

OpenPLC

Purpose

The `openplc/` component provides the simulated programmable logic controller (PLC) used as the control-system endpoint of the OT testbed.

Role in the Architecture

OpenPLC represents the field-control layer of the laboratory environment. It exposes PLC data that can be accessed through Modbus/TCP and consumed by SCADA.

Conceptually:

```
Modbus Client
Traffic Gen.    > OpenPLC  > SCADA/HMI
Attacker
```

Responsibilities

- Execute the simulated PLC control logic.
- Maintain process-related variables.
- Provide Modbus/TCP communication.
- Serve as the authoritative simulated process-control endpoint.
- Provide a target for controlled OT security experiments.

PLC Logic

The PLC logic should represent a deterministic industrial process suitable for experimentation. Changes to registers, coils, or process variables should be observable from the SCADA layer where applicable.

Testing

OpenPLC should be tested in stages:

1. Confirm the container starts successfully.
2. Confirm the PLC runtime is operational.
3. Confirm Modbus/TCP connectivity.
4. Read known PLC values with `modbus-client/`.

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5. Confirm normal background traffic from `traffic/`.
6. Validate controlled writes from the laboratory attacker.
7. Confirm corresponding SCADA changes.

Security-Lab Role

The PLC is intentionally exposed to simulated OT traffic so that communication, monitoring, and controlled security-testing workflows can be studied without using a production controller.

Related Components

- `modbus-client/`
- `traffic/`
- `attacker/`
- `scadabr/`

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